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MID SEMESTER EXAMINATION-2025

Name of the Program: B.Tech. (Bio Tech.)
 Name of the Course: Engineering Mathematics

Semester: II
 Course Code: TMA-202

Time: 90 Minutes

Maximum Marks: 50

Note: -

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
 (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Define matrix and their types with examples.	
	OR	
(b)	Find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$.	CO₁
Q2	(10 marks)	
(a)	Define singular and non-singular matrix with examples and if $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$, then find (i) AA^T , and (ii) $A^T A$.	CO₁
	OR	
(b)	Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$.	
Q3	(10 marks)	
(a)	Show that the matrix $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ satisfies the matrix equation $A^2 - 5A + 7I = O$, where I is identity matrix and O is null matrix.	CO₁
	OR	
(b)	Check the continuity of the function: $f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} \leq x < 0 \\ 3 - 2x, & 0 \leq x \leq \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$	CO₂
Q4	(10 marks)	
(a)	Find x, y, z and w given that: $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$.	CO₁
	OR	

(b)	Verify Cayley-Hamilton theorem for the following matrix and find A^{-1} : $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$	
Q5	(10 marks)	
	Find $\frac{dy}{dx}$ of the following with respect to x:	CO₂
(a)	(i) $y = \frac{x^2 + 5x + 1}{x + 2}$. (ii) $y = \frac{1 + e^x}{1 - e^x}$.	
OR		
(b)	Evaluate the following: (i) $\lim_{x \rightarrow 0} \frac{\sqrt{1 + 4x} - \sqrt{1 - 5x}}{x}$ (ii) $\lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 + x^2}}{x^2}$	